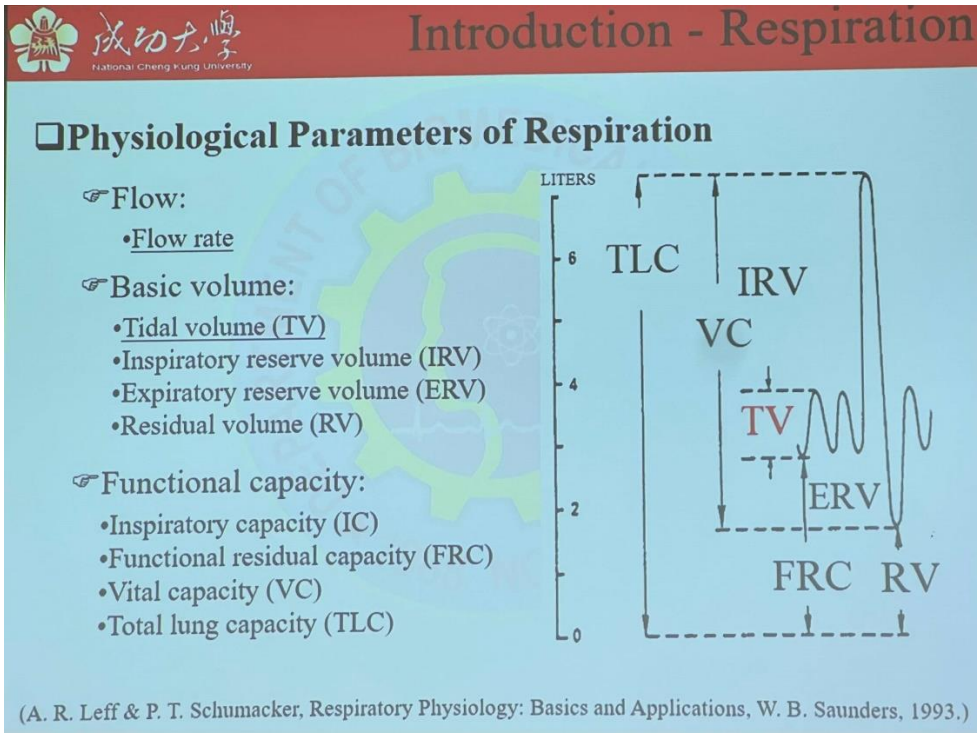
The Development and Applications of BioImpedance Technology

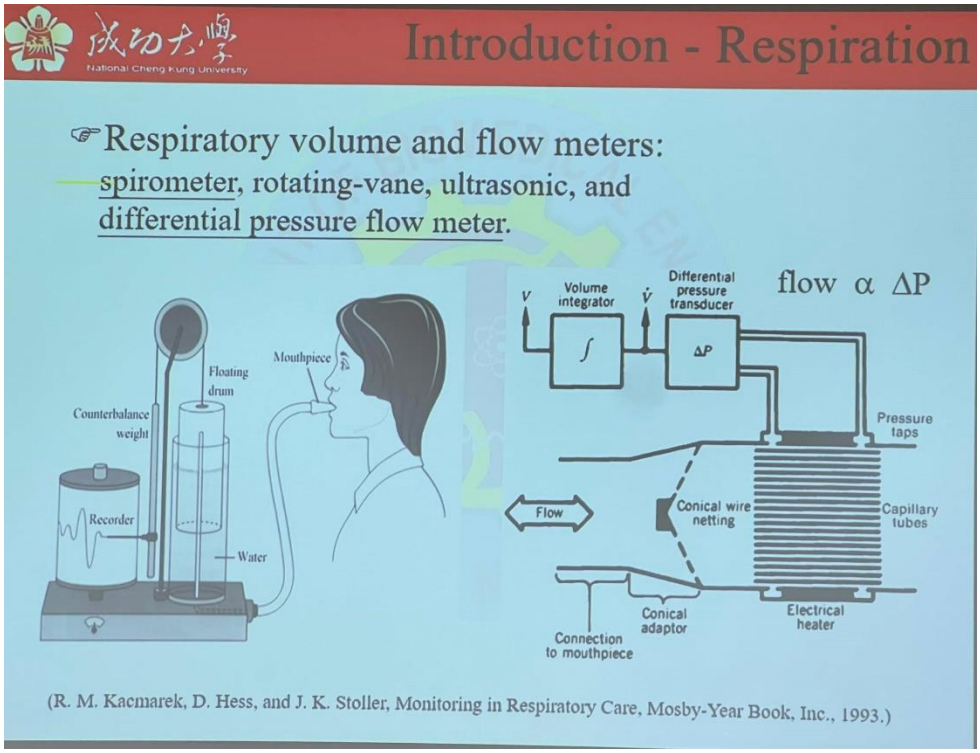
國立成功大學 鄭國順 教授

2025-03-17

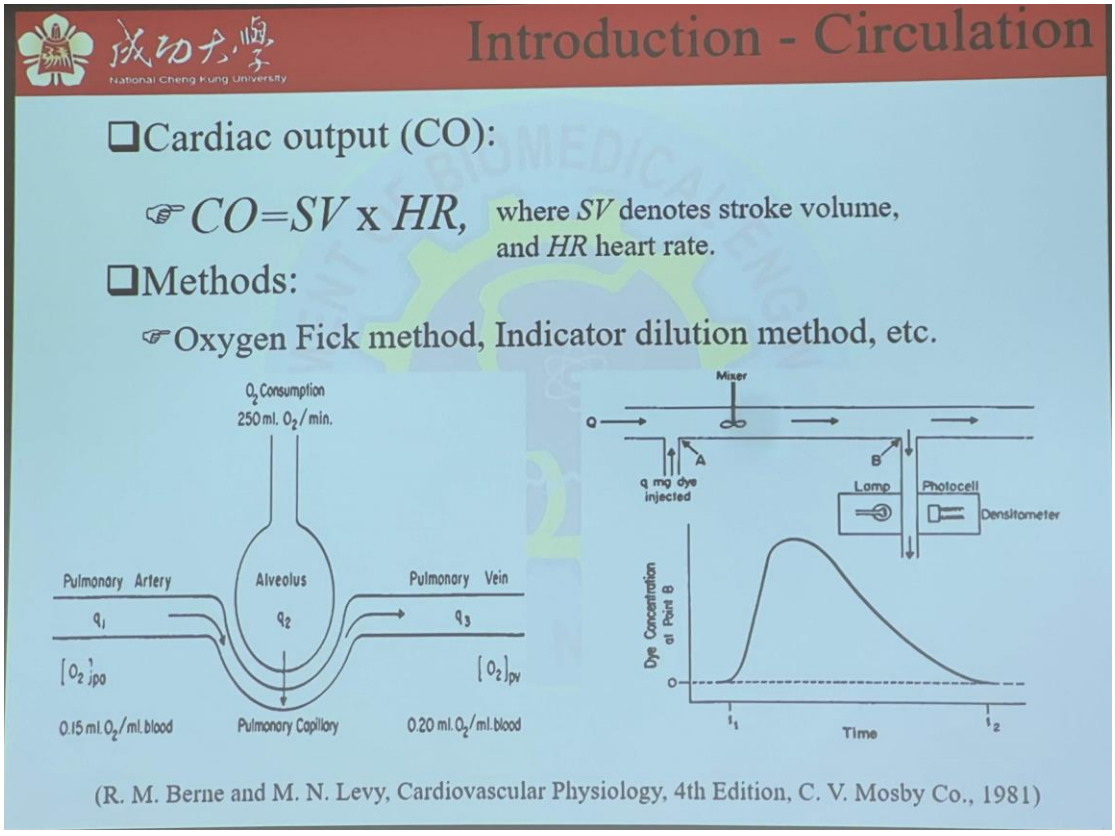
筆記：

Introduction-Respiration

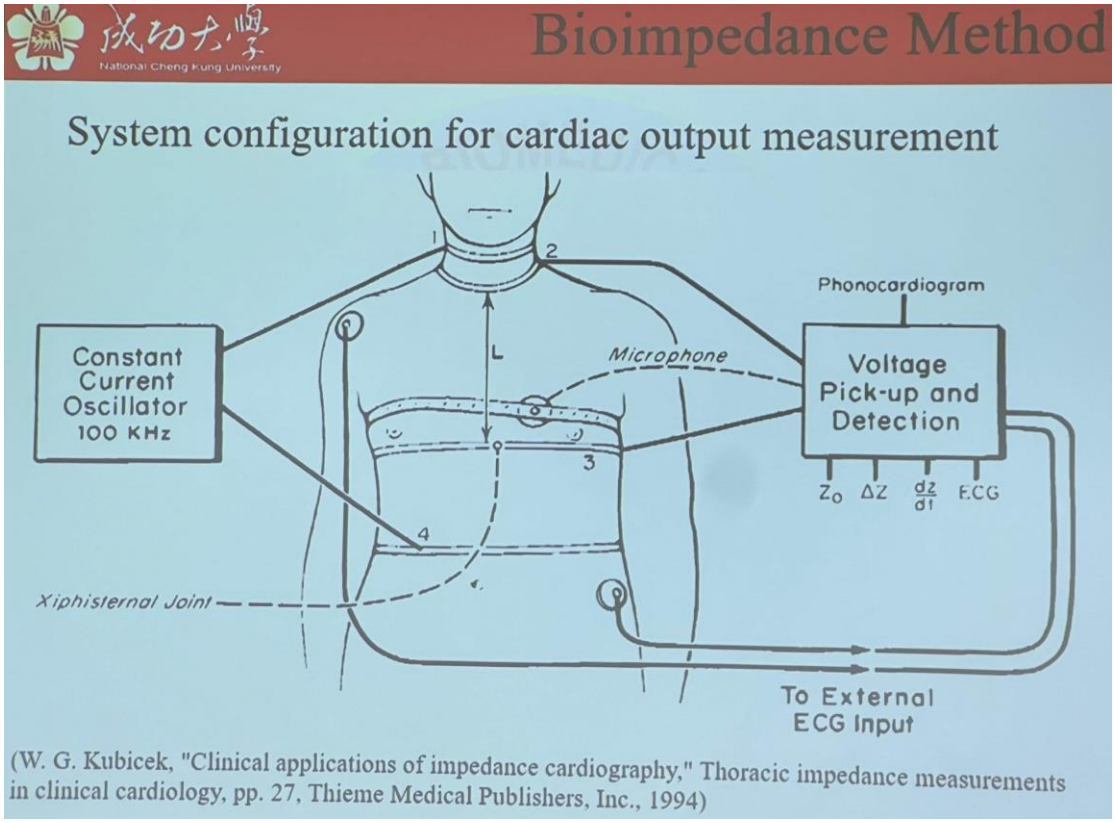


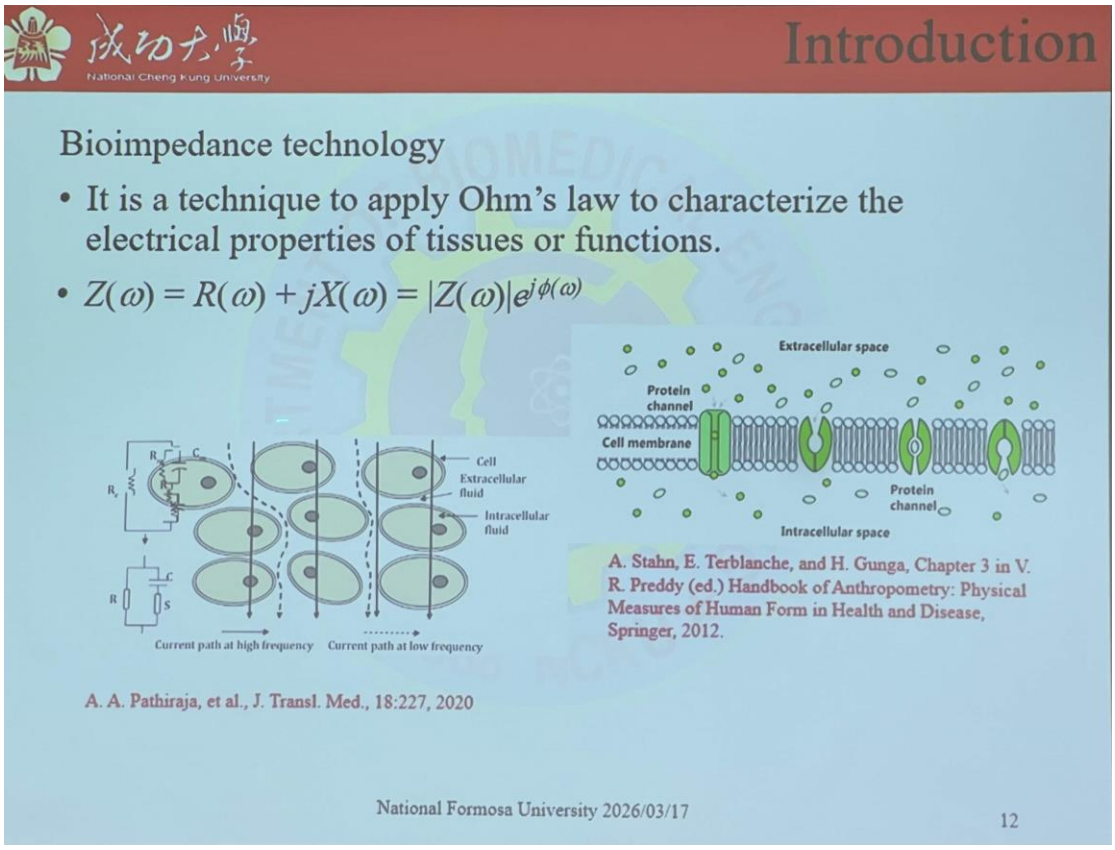


Introduction-Circulation

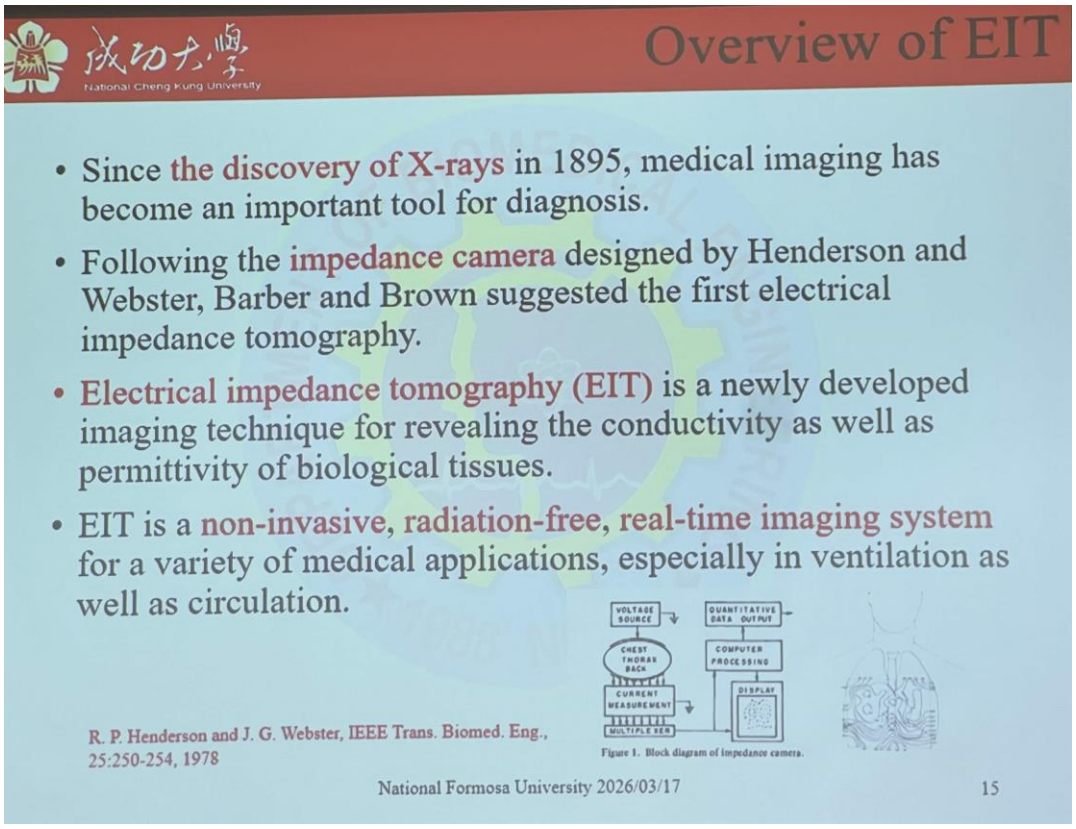


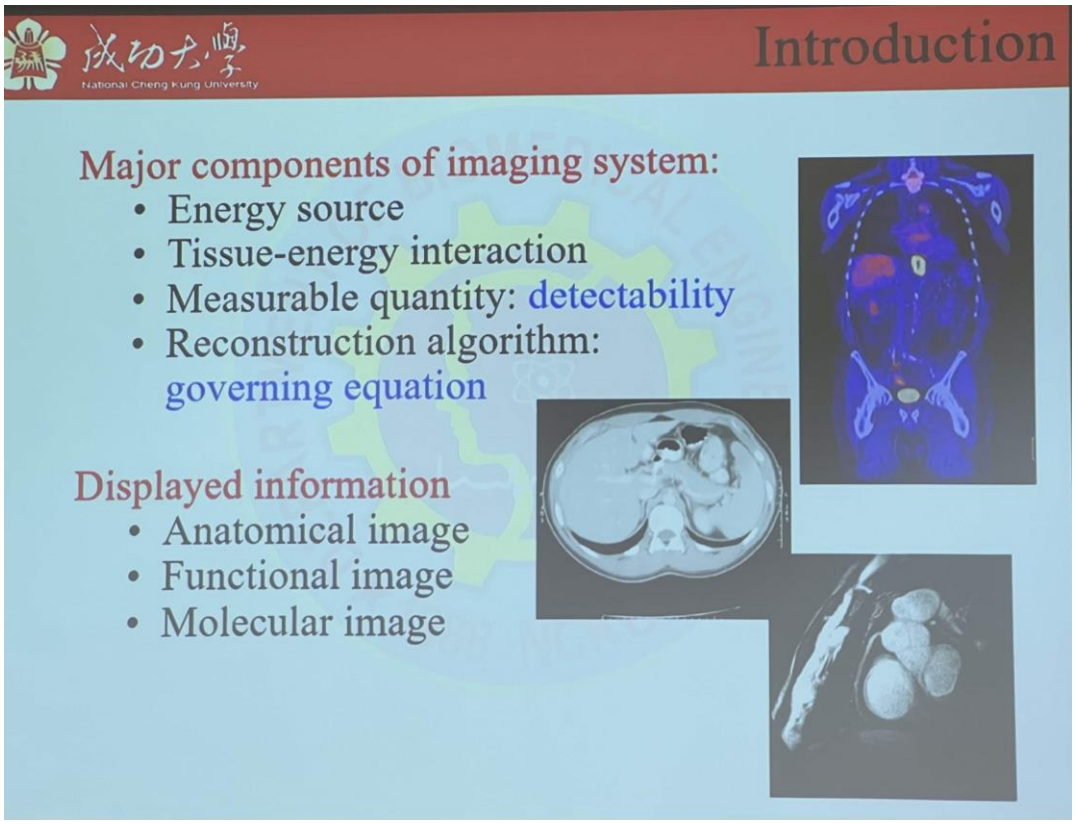
Bioimpedance Method





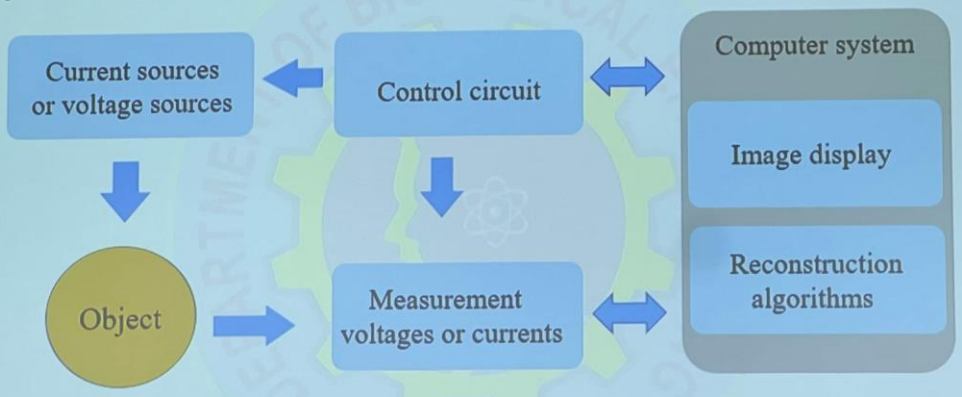
Electrical impedance tomography(EIT)





Overview of EIT

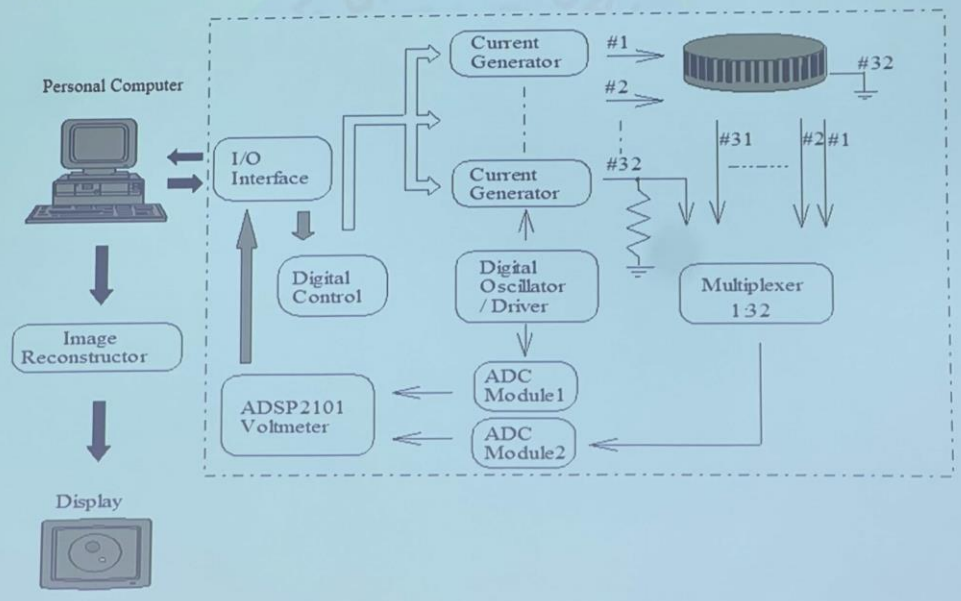
System Configuration



- Electrode/Electrode array
- Two-electrode or Four-electrode measurement
- Current or Voltage applications

Overview of EIT

System Configuration



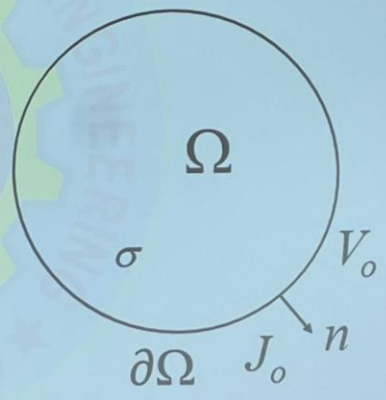
Governing Equation

Image Reconstruction Algorithms

The governing equation for EIT is the Poisson equation with boundary conditions.

$$\nabla \cdot \sigma \nabla V = 0 \text{ in } \Omega$$

$$\begin{cases} V = V_0 \\ \sigma \dfrac{\partial V}{\partial n} = J_0 \end{cases} \quad \text{on } \partial\Omega$$



where V is the voltage, J the current density, σ the conductivity, and n normal component on Ω

Forward & Inverse Problems

Image Reconstruction Algorithms

- **Forward problem**

Given internal resistivity ρ boundary current density J_o → **Forward Solver** → Find voltage V, and J

Solver: Analytic method, FEM , ...

- **Inverse problem**

Given boundary voltage V_o and current density J_o → **Inverse Solver** → Find internal resistivity

Solver: NOSER, Backprojection, GA, SA, NN, DL, ...

Forward & Inverse Problems

Image Reconstruction Algorithms

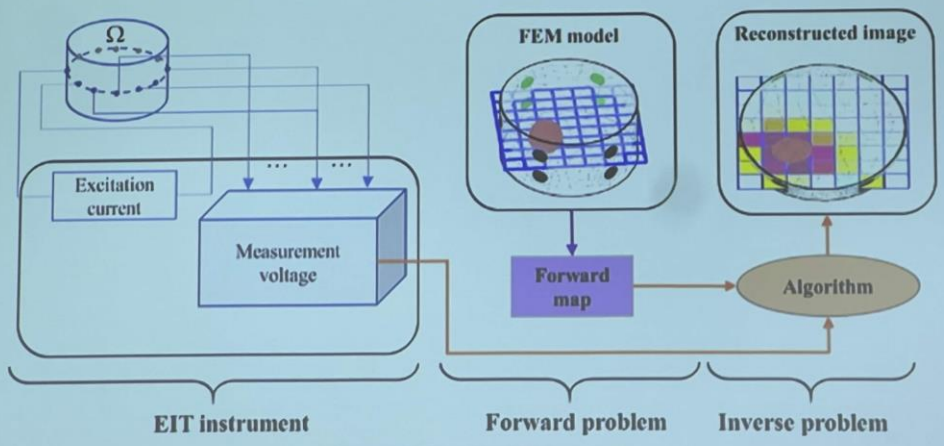


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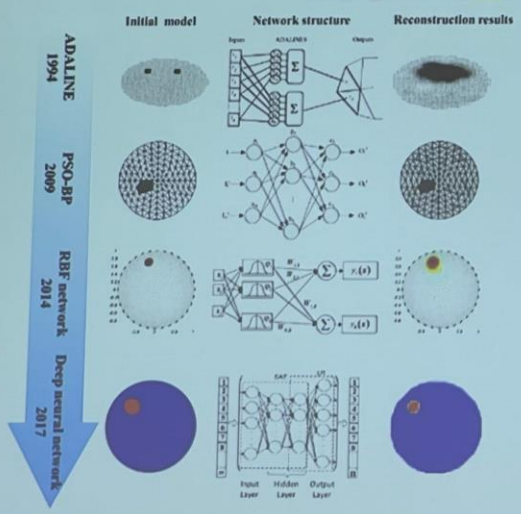
Image Reconstruction Algorithm

1. Neural network-based image reconstruction algorithms



2. Traditional image reconstruction algorithms

- Filtered backprojection
- Newton One-Step Estimation Reconstruction
- Simulated annealing
- Genetic algorithms

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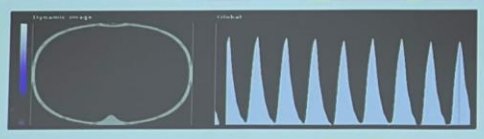
Clinical Applications of EIT

Clinical Applications in Lung Function Characterization

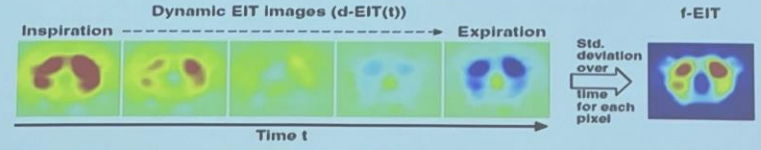
Advantages:

1. No radiation hazard
2. Non-invasive technique
3. Real-time and continuous measurement
4. Portability for bedside monitoring



Global impedance changes

http://www.draeger.com



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